

2020 AMC 10A Problem 6 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10A Problem 6. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10A Problem 6

Problem:

How many 4-digit positive integers (that is, integers between 1000 and 9999, inclusive) having only even digits are divisible by 5?

Answer Choices:

- A. 80
- B. 100
- C. 125
- D. 200
- E. 500

Solution:

The units digit, for all numbers divisible by 5, must be either 0 or 5. However, since all digits are even, the units digit must be 0.

The middle two digits can be 0, 2, 4, 6, or 8, but the thousands digit can only be 2, 4, 6, or 8 since it cannot be zero.

There is one choice for the units digit, 5 choices for each of the middle 2 digits, and 4 choices for the thousands digit, so there is a total of

$$4 \cdot 5 \cdot 5 \cdot 1 = \boxed{(B) 100} \text{ numbers.}$$

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